

DATA EXPLORATION OF IRIS DATASET

Internship Project Report

1. Introduction

Data Exploration is an important step in Data Science and Machine Learning. It helps in understanding the structure, quality, and characteristics of a dataset before applying machine learning algorithms. The Iris dataset is commonly used for learning data analysis concepts.

2. Objective

Load and explore the Iris dataset, check missing values, perform statistical analysis, visualize data, and analyze relationships among features.

3. Dataset Description

The Iris dataset contains 150 records and 4 numerical features: Sepal Length, Sepal Width, Petal Length, and Petal Width. The target classes are Setosa, Versicolor, and Virginica.

4. Tools and Libraries Used

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, and Jupyter Notebook.

5. Methodology

Loaded the dataset, explored its structure, checked for missing values, generated statistical summaries, and created visualizations such as Histograms, Boxplots, Heatmaps, Pairplots, and Countplots.

6. Results and Observations

The dataset contains 150 records. No missing values were found. Petal Length and Petal Width show strong positive correlation. Setosa is clearly distinguishable from the other species. The dataset is balanced with 50 samples per species.

7. Conclusion

The Iris dataset was successfully explored and analyzed. The dataset is clean and suitable for machine learning applications. This project demonstrates the basic process of Exploratory Data Analysis (EDA).